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SI—29—2025

FACULTY OF SCIENCE AND TECHNOLOGY

BCA (Second Year) (Fourth Semester) EXAMINATION

APRIL/MAY, 2025

(CBCS/Revised Pattern)

COMPUTER APPLICATION

Paper–BCA–403

(Relational Database Management System)

(Saturday, 12-4-2025) **Time : 2.00 p.m. to 5.00 p.m.**

Time—3 Hours *Maximum Marks—75*

N.B. :— (*i*) *All* questions are compulsory.

(*ii*) Figures to the right indicate full marks.

(*iii*) Assume suitable data, if required.

1. Write short notes on any *five* of the following (**3** marks each) : 15

(*a*) Single Row Conversion.

(*b*) TCL.

(*c*) Foreign Key.

(*d*) Sorting.

(*e*) Views.

(*f*) Subqueries.

(*g*) Cross join.

P.T.O.

2. Attempt any *three* of the following (**5** marks each) : 15

(*a*) Explain applications of RDBMS.

(*b*) Explain the advantages of RDBMS.

(*c*) Explain the characteristics of RDBMS.

(*d*) Explain object oriented data models in detail.

(*e*) Explain mapping ER model to relational mode.

3. Attempt any *three* of the following (**5** marks each) : 15

(*a*) Explain data types in SQL.

(*b*) Explain data constraints in brief.

(*c*) Explain how to manipulate data in table.

(*d*) Explain entity relationship.

(*e*) What is SQL ? Explain SQL commands.

4. Attempt any *three* of the following (**5** marks each) : 15

(*a*) Explain SQL functions in detail.

(*b*) Explain multiple row functions.

(*c*) What is sorting ? Explain ORDER BY.

(*d*) What is operator ? Explain operators in detail.

(*e*) Explain the types of subqueries.

5. Attempt any *three* of the following (**5** marks each) : 15

(*a*) Explain the declaration section in PL/SQL.

(*b*) Explain views.

(*c*) Explain join and its types.

(*d*) Discuss altering table structure.

(*e*) Explain DISTINCT.

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